

Module 1

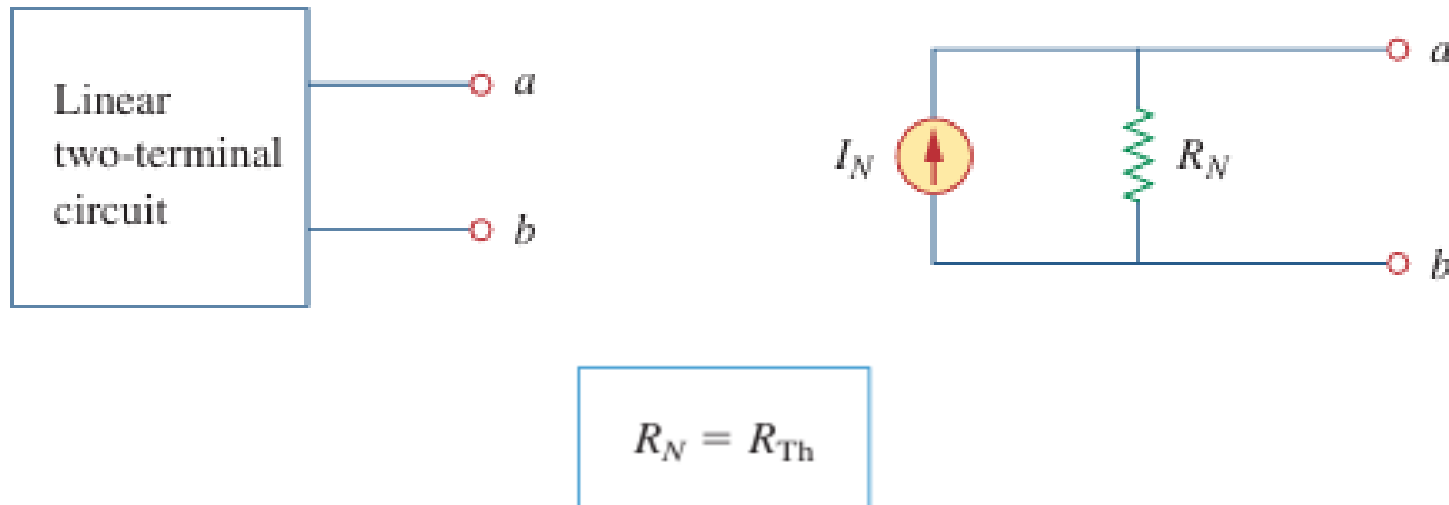
DC Circuit Analysis and Theorems

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Norton's Theorem

Norton's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N , where I_N is the short-circuit current through the terminals and R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.

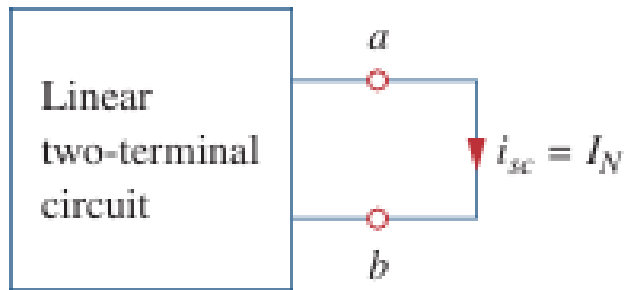


Two circuits are said to be equivalent if they have the same voltage-current relation at their terminals.

Norton's Theorem

Norton's equivalent Current I_N and Norton's resistance R_N :

1. To find the Norton current I_N , determine the short-circuit current flowing from terminal a to b .
2. R_N is the input resistance at the terminals when the independent sources are turned off. This is the **same** as R_{th} .



$$I_N = i_{sc}$$

V_{th} , R_{th} , and I_N are related according to

$$I_N = \frac{V_{Th}}{R_{Th}}$$

Dependent and independent sources are treated the same way as in Thevenin's theorem.

This is essentially source transformation.

For this reason, source transformation is often called Thevenin-Norton transformation.

Norton's Theorem

To determine the Thevenin or Norton equivalent circuit, it requires that we find:

1. The open-circuit voltage across terminals a and b.
2. The short-circuit current at terminals a and b.
3. The equivalent or input resistance at terminals a and b when all independent sources are turned off.

We can calculate any two of the three using the method that takes the least effort and use them to get the third using Ohm's law.

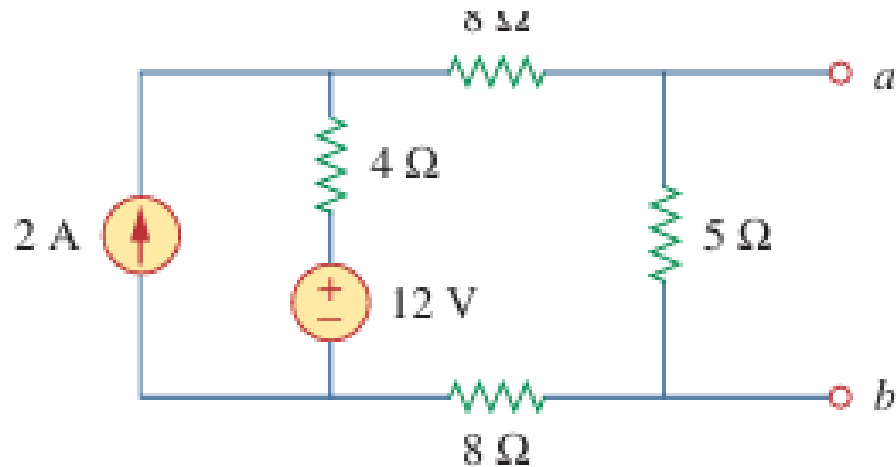
$$V_{Th} = v_{oc}$$

$$I_N = i_{sc}$$

$$R_{Th} = \frac{v_{oc}}{i_{sc}} = R_N$$

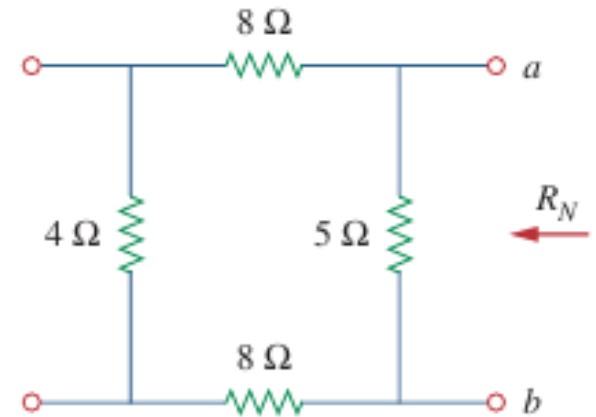
Norton's Theorem

Find the Norton equivalent circuit of the circuit below at terminals a-b.



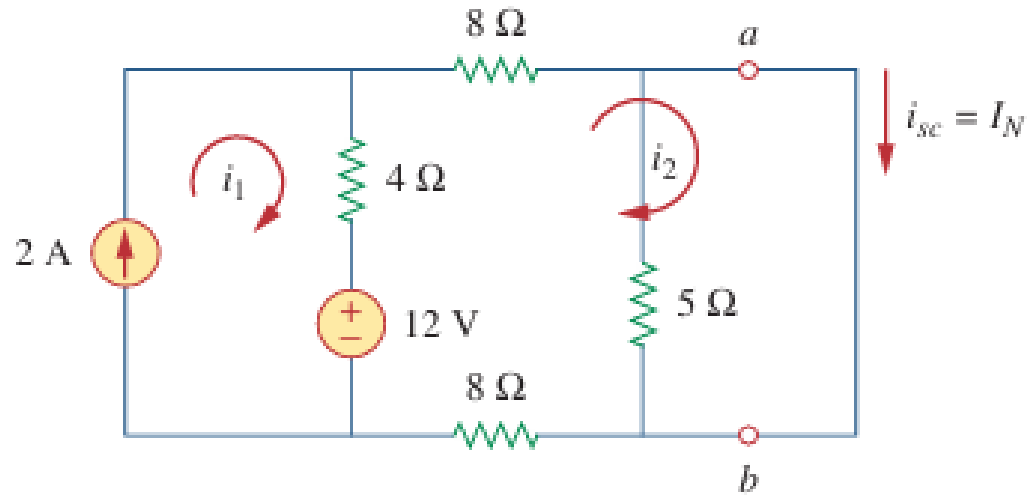
R_N

$$R_N = 5 \parallel (8 + 4 + 8) = 5 \parallel 20 = \frac{20 \times 5}{25} = 4 \Omega$$



Norton's Theorem

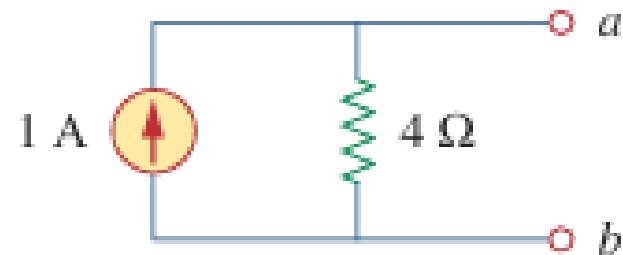
I_N



$$i_1 = 2 \text{ A}$$

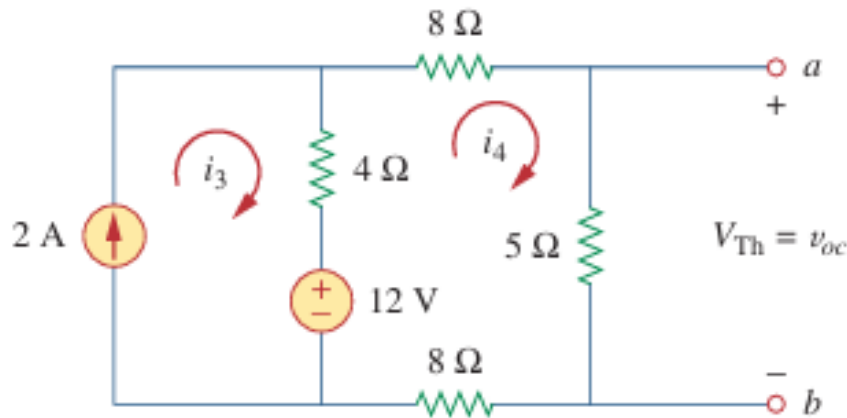
$$20i_2 - 4i_1 - 12 = 0$$

$$i_2 = 1 \text{ A} = i_{sc} = I_N$$



Norton's Theorem

Alternatively, we can also determine I_N from V_{th} .



$$I_N = \frac{V_{Th}}{R_{Th}}$$

$$i_3 = 2 \text{ A}$$

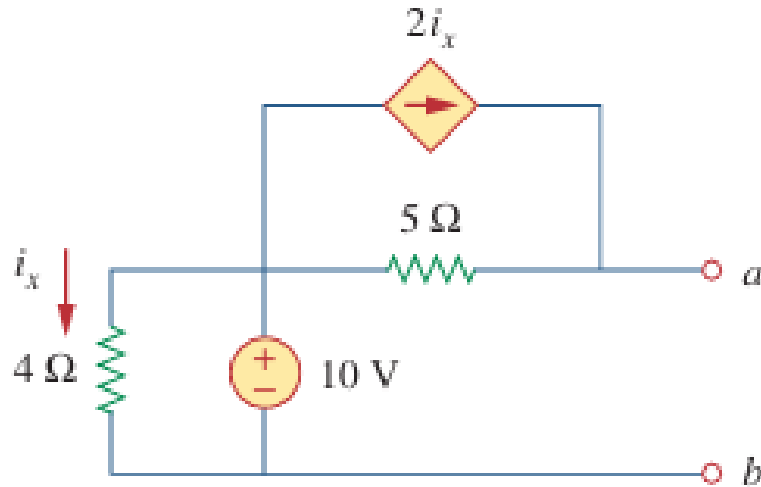
$$25i_4 - 4i_3 - 12 = 0 \quad \Rightarrow \quad i_4 = 0.8 \text{ A}$$

$$v_{oc} = V_{Th} = 5i_4 = 4 \text{ V}$$

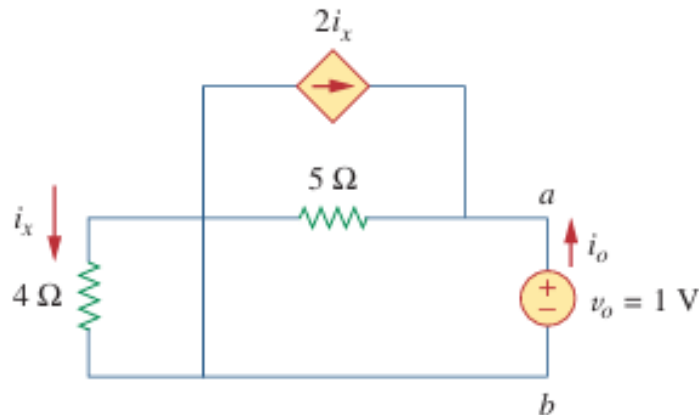
$$I_N = \frac{V_{Th}}{R_{Th}} = \frac{4}{4} = 1 \text{ A}$$

Norton's Theorem

Find the Norton's equivalent of the circuit at terminals a-b.



R_N



1. To find $R_{th} = R_N$ we set the independent source equal to zero but leave the dependent source alone.

2. However, we excite the network with a voltage source v_o connected to the terminals.

3. For easy calculation set $v_o = 1$ V.

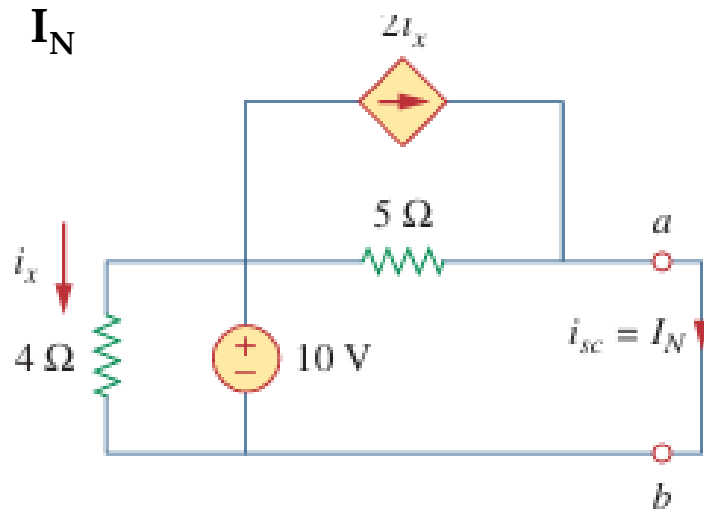
4. Let us now find the current i_o through the terminals, and then obtain $R_N = v_o/i_o$

$$i_x = 0.$$

$$i_o = \frac{1v}{5\Omega} = 0.2 \text{ A}$$

$$R_N = \frac{v_o}{i_o} = \frac{1}{0.2} = 5 \Omega$$

Norton's Theorem



$4\ \Omega$ resistor, 10 V voltage source, the $5\ \Omega$ resistor, and the dependent current source are all in parallel.

$$i_x = \frac{10}{4} = 2.5\text{ A}$$

At node **a**, KCL gives

$$i_{sc} = \frac{10}{5} + 2i_x = 2 + 2(2.5) = 7\text{ A}$$

$$I_N = 7\text{ A}$$

Next class: Maximum Power transfer Theorem